

Muhammad Sharjeel Akhtar

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EDUCATION

Brandenburgische Technische Universität (BTU), Cottbus, Germany *April 2025 – Present*
Master of Science in Computer Science (Specialization: Artificial Intelligence)

National University of Computer & Emerging Sciences (FAST), Pakistan *August 2020 – May 2024*
Bachelor of Science in Computer Science

TECHNICAL SKILLS

Programming: Python, SQL, C++, Java, JavaScript, PHP, Bash, HTML/CSS

AI & Frameworks: Transformers (HuggingFace), LangChain, LangGraph, PyTorch, Keras, Scikit-learn, Ollama

Data & Storage: FAISS, Vector Databases, PostgreSQL, MongoDB, MySQL, Pandas, NumPy, Neo4j

DevOps & Cloud: Docker, Kubernetes, AWS (RDS, EC2, Load Balancer, Auto Scaling), Git, Linux, Uvicorn

APIs & Deployment: FastAPI, Streamlit, REST APIs, Pydantic, Apache · **IIoT:** N8N, Node-RED, SCADA

Languages & Communication: English (Fluent, IELTS Certified) · German (A1, actively learning)

WORK EXPERIENCE

Pakistan Aeronautical Complex (PAC), Kamra, Pakistan *June 2023*
Engineering Internship – Avionics Production Factory (APF)

- Worked with avionics stress-testing software to validate hardware-software integration on jet systems.
- Observed and analyzed radar monitoring and observability dashboards used in real-time flight diagnostics.
- Reviewed embedded software documentation and system logs to support diagnostic accuracy verification.

Automation Play (Industry 4.0 IIoT Boot Camp), Remote *Nov 2023 – Dec 2023*
Industry 4.0 IIoT and Machine Learning Intern

- Engineered AI/ML evaluation metrics over simulated industrial data telemetry streams.
- Structured Node-RED logic pipelines to manage remote cloud backend integrations.
- Analyzed data transmission flows bridging factory devices with central SCADA monitoring setups.

CodSoft, Remote *July 2023 – Oct 2023*
Software Engineering and Web Development Intern

- Developed Python backend modules to clean, transform, and structure live data pipelines.
- Designed responsive frontend views integrated against localized web database engines.

KEY AI & ENGINEERING PROJECTS

Multi-Agent AI Research and Content Generation System (LangGraph) *Project Link* 🔗

- Designed a cyclic multi-agent workflow via LangGraph with router, research, planning, and worker nodes.
- Integrated Tavily API search loops with Pydantic layers to enforce structural schema validation.
- Created context-driven visual content generation prompts deploying Gemma 4 models locally via Ollama.

Corrective RAG (CRAG) with Self-Correction Pipeline *Project Link* 🔗

- Developed a CRAG engine scoring retrieval quality to dynamically trigger web search fallbacks.
- Indexed reference corpuses using FAISS vector spaces and Qwen3 transformer embedding models.
- Applied sentence-level filtering to reduce irrelevant token injection before generation execution.

Production MLOps Pipeline - FastAPI & Docker *Project Link* 🔗

- Engineered an asynchronous FastAPI service hosting a containerized insurance pricing ML engine.
- Configured reproducible Docker microservices with automated Docker Hub version tracking.

Scalable AWS Three-Tier Architecture Deployment *Project Link* 🔗

- Deployed three-tier AWS infrastructure (EC2, RDS, Load Balancer, Auto Scaling) ensuring high availability.

CERTIFICATIONS

[View All Verified Certificates](#) 🔗

English Language Proficiency (IELTS) · Industry 4.0 and IIoT (Automation Play) · Machine Learning and Deep Learning (DeepLearning.AI)